

```

import pandas as pd
from google.colab import drive
#create a csv file
data = {
    "Name": ["Amit", "Neha", "Raj", "Sneha", "Vikas", "Pooja", "Karan", "Anjali", "Rohit", "Meena",
            "Arjun", "Priya", "Sahil", "Kavya", "Manoj", "Ritu", "Deepak", "Simran", "Nikhil", "Asha"],

    "Branch": ["Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil",
              "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil"],

    "Age": [18, 19, 18, 19, 20, 18, 19, 20, 18, 19, 20, 18, 19, 20, 18, 19, 20, 18, 19, 20],

    "Marks": [78, 85, 67, 92, 74, 81, 69, 88, 73, 90, 65, 86, 72, 91, 68, 87, 70, 89, 75, 93],

    "Attendance": [85, 90, 80, 95, 88, 87, 82, 91, 84, 93, 78, 89, 81, 94, 79, 92, 83, 96, 86, 97]
}

# Convert to DataFrame
df = pd.DataFrame(data)

# Save as CSV
df.to_csv("students_data.csv", index=False)

print("CSV file created successfully!")
drive.mount('/content/drive')

df.to_csv('/content/drive/MyDrive/Colab Notebooks/students_data.csv', index=False)

```

CSV file created successfully!
Mounted at /content/drive

```

df = pd.read_csv("students_data.csv")
display(df)

```

	Name	Branch	Age	Marks	Attendance
0	Amit	Mechanical	18	78	85
1	Neha	Civil	19	85	90
2	Raj	Mechanical	18	67	80
3	Sneha	Civil	19	92	95
4	Vikas	Mechanical	20	74	88
5	Pooja	Civil	18	81	87
6	Karan	Mechanical	19	69	82
7	Anjali	Civil	20	88	91
8	Rohit	Mechanical	18	73	84
9	Meena	Civil	19	90	93
10	Arjun	Mechanical	20	65	78
11	Priya	Civil	18	86	89
12	Sahil	Mechanical	19	72	81
13	Kavya	Civil	20	91	94
14	Manoj	Mechanical	18	68	79
15	Ritu	Civil	19	87	92
16	Deepak	Mechanical	20	70	83
17	Simran	Civil	18	89	96
18	Nikhil	Mechanical	19	75	86
19	Asha	Civil	20	93	97

```
print(type(df))
```

```
<class 'pandas.core.frame.DataFrame'>
```



```
print(df.describe())
```

	Age	Marks	Attendance
count	20.000000	20.000000	20.000000
mean	18.950000	79.650000	87.500000
std	0.825578	9.527108	5.91608
min	18.000000	65.000000	78.000000
25%	18.000000	71.500000	82.75000
50%	19.000000	79.500000	87.50000
75%	20.000000	88.250000	92.25000
max	20.000000	93.000000	97.00000

```
print(df['Age'].mean())
```

18.95

```
print("Minimum value",df['Marks'].min())
```

Minimum value 65

```
print(df['Marks'].mean())
```

79.65

```
print(df['Marks'].max())
```

93

```
print(df['Marks'].median())
```

79.5

```
print(df['Marks'].mode())
```

```
0    65
1    67
2    68
3    69
4    70
5    72
6    73
7    74
8    75
9    78
10   81
11   85
12   86
13   87
14   88
15   89
16   90
17   91
18   92
19   93
Name: Marks, dtype: int64
```

```
print(df[['Marks','Age']].mean())
```

```
Marks    79.65
Age      18.95
dtype: float64
```

```
print(df['Age'].value_counts())
```

```
Age
18    7
19    7
20    6
Name: count, dtype: int64
```

```
print(df.count())
```

```
Name      20
Branch    20
Age       20
Marks     20
```



```
Attendance    20
dtype: int64
```

```
print(df['Marks'].sum())
```

```
1593
```

```
print(df.groupby('Branch')['Marks'].sum())
```

```
Branch
Civil      882
Mechanical 711
Name: Marks, dtype: int64
```

```
print(df.groupby('Age')['Marks'].sum())
```

```
Age
18    542
19    570
20    481
Name: Marks, dtype: int64
```

```
print(df.groupby(['Branch', 'Age'])['Marks'].mean())
```

```
Branch    Age
Civil     18    85.333333
          19    88.500000
          20    90.666667
Mechanical 18    71.500000
          19    72.000000
          20    69.666667
Name: Marks, dtype: float64
```

```
print(df.groupby('Branch')['Marks'].agg(['mean', 'max', 'min']))
```

```
          mean  max  min
Branch
Civil         88.2   93   81
Mechanical    71.1   78   65
```

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Re-create data to ensure it exists
```

```
data = {
    "Name": ["Amit", "Neha", "Raj", "Sneha", "Vikas", "Pooja", "Karan", "Anjali", "Rohit", "Meena",
             "Arjun", "Priya", "Sahil", "Kavya", "Manoj", "Ritu", "Deepak", "Simran", "Nikhil", "Asha"],
    "Branch": ["Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil",
               "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil", "Mechanical", "Civil"],
    "Age": [18, 19, 18, 19, 20, 18, 19, 20, 18, 19, 20, 18, 19, 20, 18, 19, 20, 18, 19, 20],
    "Marks": [78, 85, 67, 92, 74, 81, 69, 88, 73, 90, 65, 86, 72, 91, 68, 87, 70, 89, 75, 93],
    "Attendance": [85, 90, 80, 95, 88, 87, 82, 91, 84, 93, 78, 89, 81, 94, 79, 92, 83, 96, 86, 97]
}
```

```
df = pd.DataFrame(data)
```

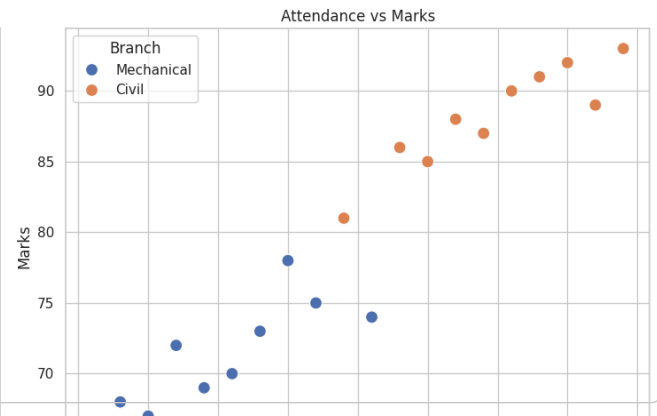
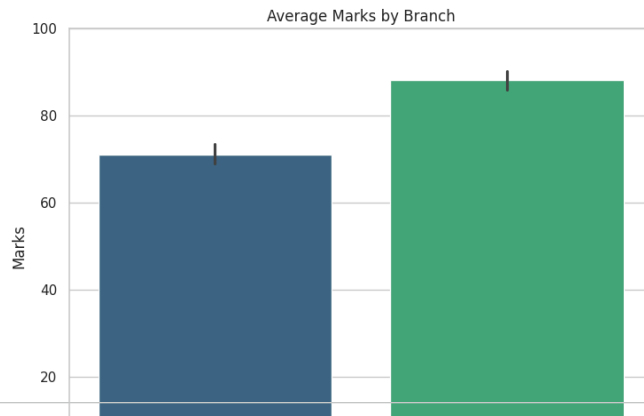
```
# Set the visual style
sns.set_theme(style="whitegrid")
```

```
# Create a figure with two subplots
fig, axes = plt.subplots(1, 2, figsize=(15, 6))
```

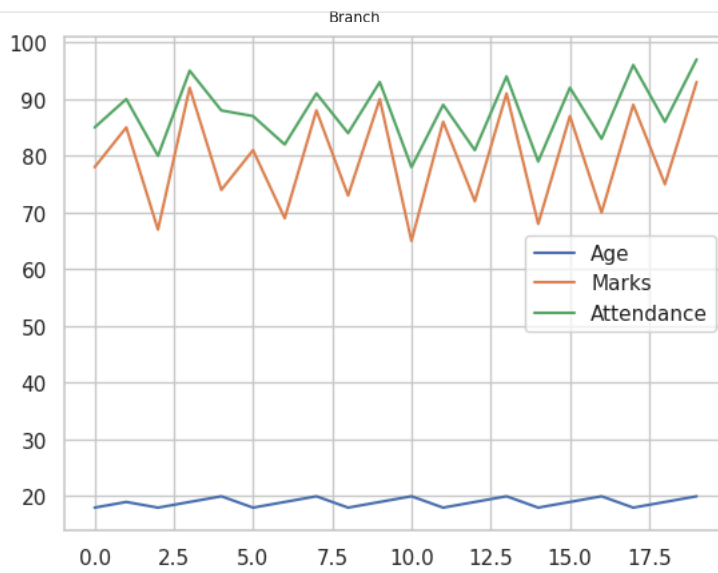
```
# Plot 1: Average Marks by Branch
sns.barplot(x='Branch', y='Marks', data=df, ax=axes[0], palette='viridis', hue='Branch', legend=False)
axes[0].set_title('Average Marks by Branch')
axes[0].set_ylim(0, 100)
```

```
# Plot 2: Attendance vs Marks
sns.scatterplot(x='Attendance', y='Marks', hue='Branch', data=df, ax=axes[1], s=100)
axes[1].set_title('Attendance vs Marks')
```

```
plt.tight_layout()
plt.show()
```

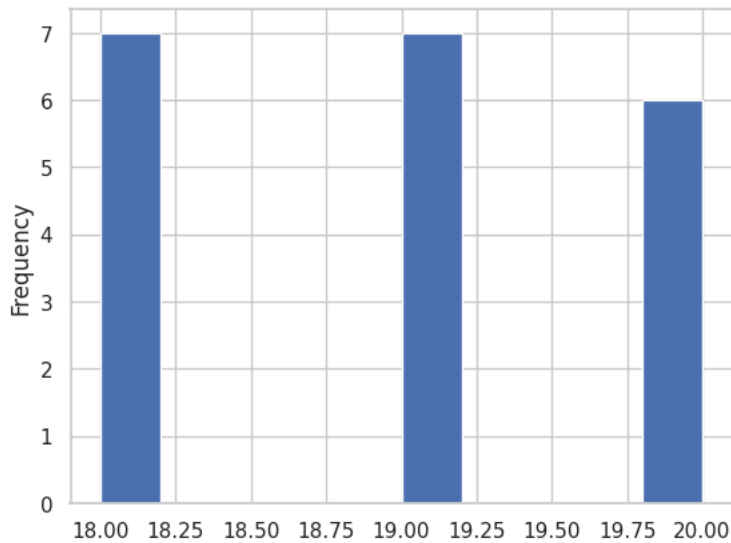


```
df.plot()  
plt.show()
```

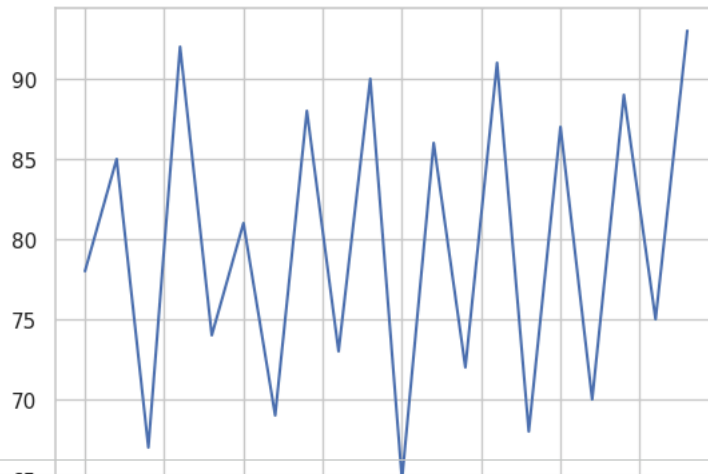


Attendance

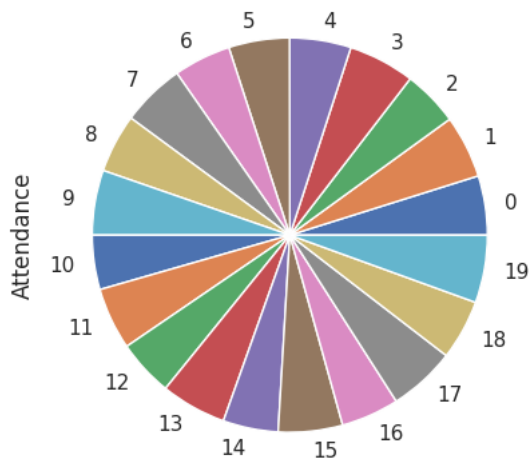
```
df["Age"].plot(kind = 'hist')  
plt.show()
```



```
df["Marks"].plot(kind = 'line')  
plt.show()
```



```
df["Attendance"].plot(kind = 'pie')  
plt.show()
```



```
import matplotlib.pyplot as plt  
  
# Correcting the syntax to use column names as strings  
df.plot(x='Marks', y='Age', kind='hist')  
plt.title('Age vs Marks Scatter Plot')  
plt.show()
```